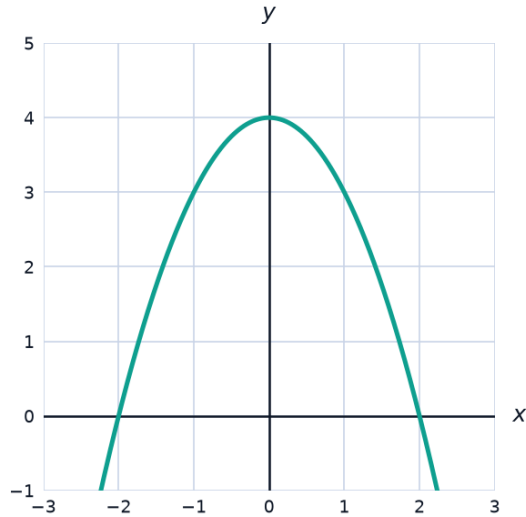


Area Between Curves



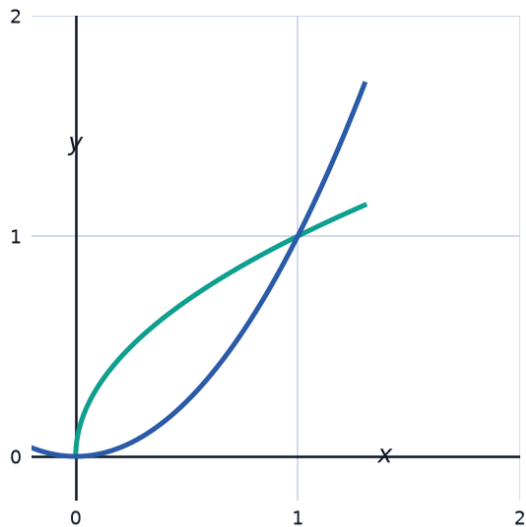
Area between a curve and the x-axis

- 1 Find the area of the region bounded by $y = 4 - x^2$ and the x-axis.



Area between two curves (vertical strips)

- 2 Find the area between $y = x^2$ and $y = x + 2$.
- 3 Find the area between $y = \sqrt{x}$ and $y = x^2$ on $[0, 1]$.



Regions requiring horizontal strips

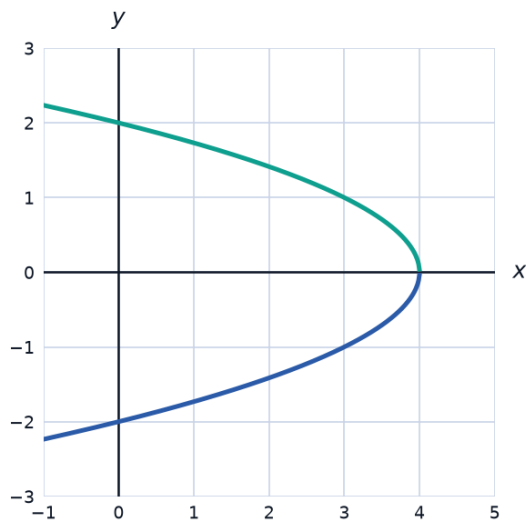
- 4 Find the area of the region bounded by $x = y^2$ and $x = y + 2$, integrating with respect to y .
-

Regions with multiple intersection points

- 5 Find the total area between $y = \sin x$ and $y = \cos x$ on $[0, \pi]$, accounting for the curves crossing at $x = \frac{\pi}{4}$.
-

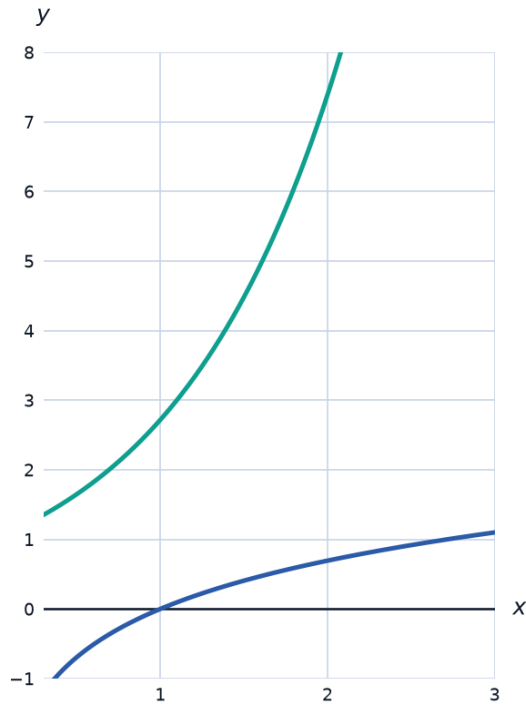
Area using the more efficient variable of integration

- 6 Find the area enclosed by $y^2 = 4 - x$ and $x = 0$, choosing the more convenient variable of integration.



Setting up but not evaluating (AP-style)

- 7 Set up (but do not evaluate) the integral expression for the area of the region enclosed by $y = e^x$, $y = \ln x$, and the lines $x = 1$ and $x = 2$.



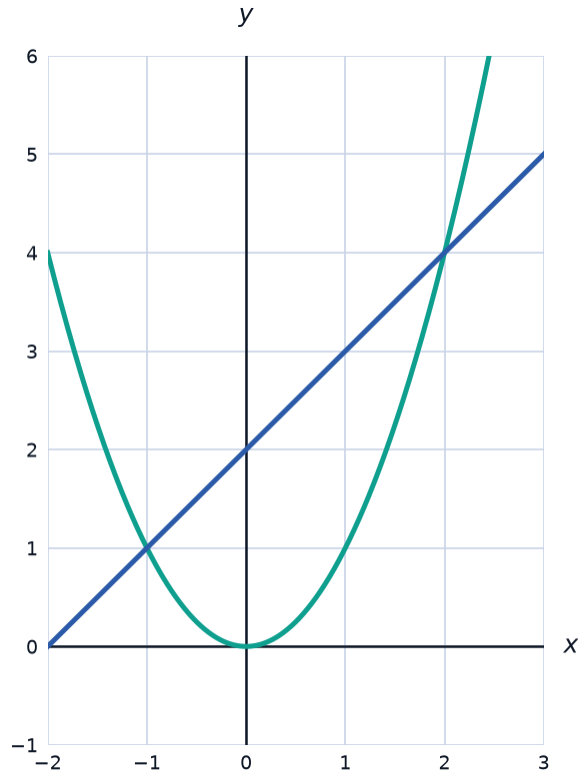
- 8 Set up (but do not evaluate) the integral expression for the area between $y = x^3 - 4x$ and the x -axis on $[-2, 2]$, accounting for the curve crossing the axis at $x = -2, 0, 2$.

Area between curves and a table of values

- 9 Functions f and g satisfy $f(x) \geq g(x)$ for all x in $[0, 6]$, with selected values given: $f(0) = 10$, $f(2) = 8$, $f(4) = 9$, $f(6) = 12$ and $g(0) = 3$, $g(2) = 4$, $g(4) = 5$, $g(6) = 6$. Use the Trapezoidal Rule with these 3 equal subintervals to approximate the area between the curves, $\int_0^6 [f(x) - g(x)] dx$.

Visualizing area between two curves

- 10 The graph shows $y = x^2$ and $y = x + 2$ from an earlier problem, together with the region between them.



A capstone area problem

- 11 Find the area of the region bounded by $y = x^3 - x$ and $y = 0$ on $[-1, 1]$, accounting for the sign change at $x = 0$.

Area between an exponential curve and a line

- 12 Find the area between $y = e^x$ and $y = x + 1$ on $[0, 2]$ (note $e^x \geq x + 1$ for all x , tangent only at $x = 0$).

An applied area-between-curves problem

- 13 Two competing companies' profit rates (in thousands of dollars per year) are $P_1(t) = 50 - 2t$ and $P_2(t) = 30 + t$ over the first 10 years. Find the total difference in cumulative profit between the two companies over this period, using the area between the curves.

Multiple choice — easy 15

14 Find the area between $y = x^2$ and $y = -8x - 12$.

a. $\frac{35}{3}$

b. $\frac{32}{3}$

c. 8

d. $\frac{64}{3}$

15 Find the area between $y = x^2$ and $y = 7x - 10$.

a. 9

b. 5

c. $\frac{9}{2}$

d. $\frac{11}{2}$

16 Find the area between $y = x^2$ and $y = 10x - 16$.

a. 37

b. 72

c. 18

d. 36

17 Find the area between $y = x^2$ and $y = -x + 6$.

a. $\frac{25}{2}$

b. $\frac{131}{6}$

c. $\frac{125}{3}$

d. $\frac{125}{6}$

18 Find the area between $y = x^2$ and $y = -4x - 3$.

a. $\frac{4}{3}$

b. 2

c. $\frac{8}{3}$

d. $\frac{7}{3}$

19 Find the area between $y = x^2$ and $y = -2x$.

a. 2

b. $\frac{4}{3}$

c. $\frac{8}{3}$

d. $\frac{7}{3}$

20 Find the area between $y = x^2$ and $y = 7x - 6$.

a. $\frac{25}{2}$

b. $\frac{131}{6}$

c. $\frac{125}{6}$

d. $\frac{125}{3}$

21 Find the area between $y = x^2$ and $y = 3x$.

a. 9

b. $\frac{11}{2}$

c. $\frac{9}{2}$

d. 4

22 Find the area between $y = x^2$ and $y = 4x + 5$.

a. 72

b. 36

c. 37

d. 18

23 Find the area between $y = x^2$ and $y = 5x - 6$.

a. $\frac{1}{2}$

b. $\frac{1}{6}$

c. $\frac{1}{3}$

d. $\frac{7}{6}$

24 Find the area between $y = x^2$ and $y = -4x + 5$.

a. 18

b. 37

c. 36

d. 72

25 Find the area between $y = x^2$ and $y = x + 2$.

a. $\frac{9}{2}$

b. 9

c. $\frac{11}{2}$

d. 5

- 26 Find the area between $y = x^2$ and $y = 6x$.
- a. 72
 - b. 18
 - c. 37
 - d. 36
- 27 Find the area between $y = x^2$ and $y = -10x - 24$.
- a. $\frac{8}{3}$
 - b. $\frac{7}{3}$
 - c. 2
 - d. $\frac{4}{3}$
- 28 Find the area between $y = x^2$ and $y = 9x - 14$.
- a. $\frac{25}{2}$
 - b. $\frac{125}{6}$
 - c. $\frac{131}{6}$
 - d. $\frac{125}{3}$
-

Multiple choice — medium 15

- 29 Find the area between $y = x^2$ and $y = x^7$ on $[0, 1]$.
- a. $\frac{5}{24}$
 - b. $\frac{11}{24}$
 - c. $\frac{5}{14}$
 - d. $-\frac{5}{24}$

30 Find the area between $y = x^4$ and $y = x^9$ on $[0, 1]$.

a. $-\frac{1}{10}$

b. $\frac{3}{10}$

c. $\frac{1}{10}$

d. $\frac{5}{36}$

31 Find the area between $y = x$ and $y = x^6$ on $[0, 1]$.

a. $\frac{5}{6}$

b. $\frac{9}{14}$

c. $\frac{5}{14}$

d. $-\frac{5}{14}$

32 Find the area between $y = x^2$ and $y = x^3$ on $[0, 1]$.

a. $\frac{1}{12}$

b. $\frac{7}{12}$

c. $\frac{1}{6}$

d. $-\frac{1}{12}$

33 Find the area between $y = x$ and $y = x^5$ on $[0, 1]$.

a. $\frac{1}{3}$

b. $-\frac{1}{3}$

c. $\frac{2}{3}$

d. $\frac{4}{5}$

34 Find the area between $y = x^3$ and $y = x^4$ on $[0, 1]$.

a. $\frac{9}{20}$

b. $-\frac{1}{20}$

c. $\frac{1}{12}$

d. $\frac{1}{20}$

35 Find the area between $y = x^4$ and $y = x^6$ on $[0, 1]$.

a. $\frac{12}{35}$

b. $-\frac{2}{35}$

c. $\frac{1}{12}$

d. $\frac{2}{35}$

36 Find the area between $y = x^4$ and $y = x^5$ on $[0, 1]$.

a. $\frac{1}{30}$

b. $-\frac{1}{30}$

c. $\frac{1}{20}$

d. $\frac{11}{30}$

37 Find the area between $y = x^2$ and $y = x^4$ on $[0, 1]$.

a. $\frac{8}{15}$

b. $\frac{2}{15}$

c. $\frac{1}{4}$

d. $-\frac{2}{15}$

38 Find the area between $y = x^3$ and $y = x^5$ on $[0, 1]$.

a. $\frac{1}{12}$

b. $\frac{5}{12}$

c. $\frac{2}{15}$

d. $-\frac{1}{12}$

39 Find the area between $y = x^3$ and $y = x^6$ on $[0, 1]$.

a. $\frac{11}{28}$

b. $\frac{1}{6}$

c. $\frac{3}{28}$

d. $-\frac{3}{28}$

40 Find the area between $y = x$ and $y = x^3$ on $[0, 1]$.

a. $-\frac{1}{4}$

b. $\frac{3}{4}$

c. $\frac{2}{3}$

d. $\frac{1}{4}$

41 Find the area between $y = x$ and $y = x^4$ on $[0, 1]$.

a. $\frac{3}{10}$

b. $\frac{3}{4}$

c. $\frac{7}{10}$

d. $-\frac{3}{10}$

42 Find the area between $y = x^4$ and $y = x^7$ on $[0, 1]$.

- a. $\frac{13}{40}$
- b. $-\frac{3}{40}$
- c. $\frac{3}{28}$
- d. $\frac{3}{40}$

43 Find the area between $y = x^2$ and $y = x^5$ on $[0, 1]$.

- a. $\frac{1}{2}$
- b. $-\frac{1}{6}$
- c. $\frac{1}{6}$
- d. $\frac{3}{10}$

Multiple choice — hard 15

44 Find the total area between $y = x^3 - 36x$ and the x -axis on $[-6, 6]$, accounting for the sign change at $x = -6, 0, 6$.

- a. 324
- b. 654
- c. 108
- d. 648

45 Find the total area between $y = x^3 - 9x$ and the x -axis on $[-3, 3]$, accounting for the sign change at $x = -3, 0, 3$.

- a. $\frac{87}{2}$
- b. $\frac{81}{2}$
- c. $\frac{27}{2}$
- d. $\frac{81}{4}$

- 46 Find the total area between $y = x^3 - 16x$ and the x -axis on $[-4, 4]$, accounting for the sign change at $x = -4, 0, 4$.
- a. 132
 - b. 64
 - c. 128
 - d. 32
- 47 Find the total area between $y = x^3 - 4x$ and the x -axis on $[-2, 2]$, accounting for the sign change at $x = -2, 0, 2$.
- a. 4
 - b. $\frac{17}{2}$
 - c. 10
 - d. 8
- 48 Find the total area between $y = x^3 - 25x$ and the x -axis on $[-5, 5]$, accounting for the sign change at $x = -5, 0, 5$.
- a. $\frac{635}{2}$
 - b. $\frac{125}{2}$
 - c. $\frac{625}{2}$
 - d. $\frac{625}{4}$

- 49 Functions f and g satisfy $f(x) \geq g(x)$ on $[0, 3]$, with $h(x) = f(x) - g(x)$ given by $h(0) = 8$, $h(1) = 15$, $h(2) = 6$, $h(3) = 10$. Use the Trapezoidal Rule with these 3 equal subintervals (width 1) to approximate the area between the curves.
- a. 39
 - b. 29
 - c. 31
 - d. 30
- 50 Functions f and g satisfy $f(x) \geq g(x)$ on $[0, 12]$, with $h(x) = f(x) - g(x)$ given by $h(0) = 8$, $h(4) = 9$, $h(8) = 14$, $h(12) = 2$. Use the Trapezoidal Rule with these 3 equal subintervals (width 4) to approximate the area between the curves.
- a. 112
 - b. 132
 - c. 116
 - d. 108
- 51 Functions f and g satisfy $f(x) \geq g(x)$ on $[0, 6]$, with $h(x) = f(x) - g(x)$ given by $h(0) = 2$, $h(2) = 12$, $h(4) = 13$, $h(6) = 2$. Use the Trapezoidal Rule with these 3 equal subintervals (width 2) to approximate the area between the curves.
- a. 58
 - b. 56
 - c. 52
 - d. 54

- 52 Functions f and g satisfy $f(x) \geq g(x)$ on $[0, 3]$, with $h(x) = f(x) - g(x)$ given by $h(0) = 8$, $h(1) = 14$, $h(2) = 6$, $h(3) = 6$. Use the Trapezoidal Rule with these 3 equal subintervals (width 1) to approximate the area between the curves.
- a. 26
 - b. 34
 - c. 28
 - d. 27
- 53 Functions f and g satisfy $f(x) \geq g(x)$ on $[0, 12]$, with $h(x) = f(x) - g(x)$ given by $h(0) = 10$, $h(4) = 15$, $h(8) = 8$, $h(12) = 8$. Use the Trapezoidal Rule with these 3 equal subintervals (width 4) to approximate the area between the curves.
- a. 128
 - b. 132
 - c. 124
 - d. 164
- 54 Find the area between $y = e^{-2x}$ and $y = -2x + 1$ on $[0, 1]$ (note $e^{-2x} \geq -2x + 1$ for all x , tangent only at $x = 0$), rounded to two decimal places.
- a. 0.43
 - b. 0.52
 - c. 0.35
 - d. 1.43

- 55 Find the area between $y = e^{2x}$ and $y = 2x + 1$ on $[0, 2]$ (note $e^{2x} \geq 2x + 1$ for all x , tangent only at $x = 0$), rounded to two decimal places.
- a. 21.8
 - b. 24.96
 - c. 16.64
 - d. 20.8
- 56 Find the area between $y = e^{-x}$ and $y = -x + 1$ on $[0, 3]$ (note $e^{-x} \geq -x + 1$ for all x , tangent only at $x = 0$), rounded to two decimal places.
- a. 1.96
 - b. 2.45
 - c. 2.94
 - d. 3.45
- 57 Find the area between $y = e^{2x}$ and $y = 2x + 1$ on $[0, 1]$ (note $e^{2x} \geq 2x + 1$ for all x , tangent only at $x = 0$), rounded to two decimal places.
- a. 2.19
 - b. 1.43
 - c. 0.96
 - d. 1.19
- 58 Find the area between $y = e^x$ and $y = x + 1$ on $[0, 3]$ (note $e^x \geq x + 1$ for all x , tangent only at $x = 0$), rounded to two decimal places.
- a. 12.59
 - b. 13.9
 - c. 9.27
 - d. 11.59